3 Dry thermodynamics: convective instability

Reading for Day 3: Pierrehumbert Principles of Planetary Climate, Section 2.4.

3.1 Convective (in)stability

3.1.1 Adiabatic lapse rate: activity

Please get in groups of two or three to work on the following activity to derive the adiabatic lapse rate dT/dp as a function of ρ and c_p.

1. One can derive the variation in temperature with pressure or height (i.e., the "lapse rate") of a convecting (adiabatic) atmosphere from the first law of thermodynamics (Equation 2.20). With your partner(s), substitute in our relationship for enthalpy changes with time from Equation (2.26) into Equation (2.20) to derive an expression for the lapse rate in pressure coordinates dT/dp.

2. Go on to derive the lapse rate in $d\ln T/d\ln p$ space and the lapse rate in height coordinates, dT/dz. For the latter, you will need to assume hydrostatic balance.

3.1.2 Convective (in)stability criteria

Atmospheres with temperature profiles that decrease more sharply with height than the adiabatic profile are unstable to convection. This is because if we perturb a parcel at the same density as the surroundings, the parcel's density will change according to the adiabatic relation while keeping a pressure close to that of the surroundings. If the density is less than the environmental density for upward displacements (or greater for downward displacements), then the parcel will accelerate away from its initial position – this will initiate convection. Figure 3.1 shows a schematic of convectively unstable and stable temperature profiles compared to the adiabatic lapse rate. If the parcel is hotter than the surrounding

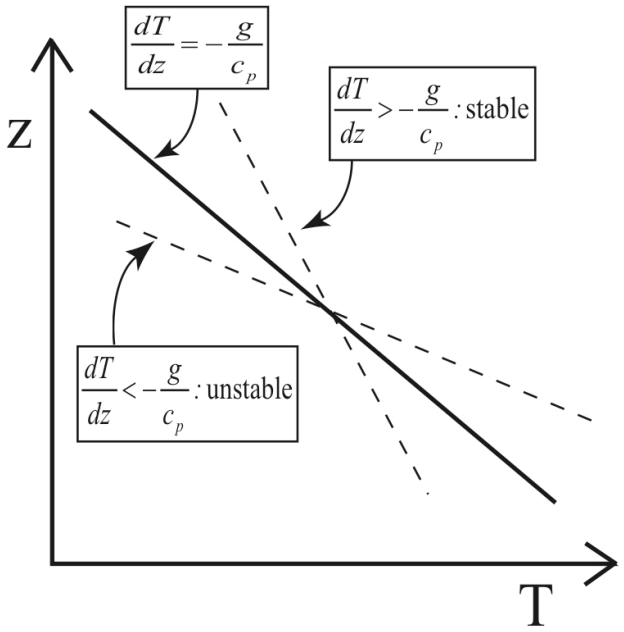


Figure 3.1: Schematic showing the convective lapse rate along with convectively unstable and stable temperature profiles. Figure credit: Adam Showman.

air than it will be less dense (and if it is cooler, then it will be more dense). The atmosphere will then be unstable to convection if temperature decreases with height faster than the adiabatic lapse rate $-g/c_p$ and stable if temperature decreases with height more slowly than the adiabatic lapse rate. Thus, we can express the convective stability of the atmosphere by comparing the environmental lapse rate to the adiabatic lapse rate as follows:

$$\begin{aligned}
\frac{dT}{dz} &< -\frac{g}{c_p} \text{ unstable,} \\
\frac{dT}{dz} &= -\frac{g}{c_p} \text{ marginally stable,} \\
\frac{dT}{dz} &> -\frac{g}{c_p} \text{ stable.}
\end{aligned} \tag{3.1}$$

Atmospheres that are forced to be unstable will adjust via convection to have lapse rates that are approximately stable, with $dT/dz \approx -g/c_p$. In general, planetary atmospheres are stable to convection except at deep levels, either near the surface for rocky planets or within the deep envelope and interior in gas giant planets.

3.1.3 Potential temperature

Note that potential temperature is a very useful quantity to determine stability of a planetary atmosphere. The potential temperature is defined such that

$$d\ln\theta \equiv d\ln T - \frac{R}{c_p} d\ln p, \tag{3.2}$$

and is equivalent to entropy in an atmospheric context[15]. For constant $R/c_p \equiv \kappa$,

$$\theta = T\left(\frac{p_o}{p}\right)^{\kappa}. \tag{3.3}$$

The potential temperature θ is thus the temperature that a parcel of dry air at pressure p and temperature T would have it if were expanded or compressed adiabatically to a reference pressure[16] p_0. We can further relate the potential temperature variation with height $d\theta/dz$ to the lapse rate dT/dz as

$$\begin{aligned}
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\frac{dT}{dz} - \frac{\kappa}{p}\frac{dp}{dz}, \\
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\frac{dT}{dz} + \frac{\kappa}{p}\rho g, \\
\frac{1}{\theta}\frac{d\theta}{dz} &= \frac{1}{T}\left(\frac{dT}{dz} + \frac{g}{c_p}\right),
\end{aligned} \tag{3.4}$$

where we have used hydrostatic equilibrium in step 2 and the ideal gas law in step 3. Note that this implies that if $d\theta/dz = 0$, the lapse rate $dT/dz = -g/c_p$ (i.e., a dry adiabat). Thus,

[15] See *Principles* Section 2.3 for a derivation of this from the second law of thermodynamics.

[16] For Earth, this reference pressure is typically taken to be 1 bar.

we can write an equivalent expression to Equation (3.1) using potential temperature:

$$\begin{aligned} \frac{d\theta}{dz} &< 0 \text{ unstable}, \\ \frac{d\theta}{dz} &= 0 \text{ marginally stable}, \\ \frac{d\theta}{dz} &> 0 \text{ stable}. \end{aligned} \tag{3.5}$$

Figure 3.2 shows an equivalent schematic to Figure 3.1 but now using potential temperature instead of temperature. An author's note is that I find potential temperature to be a

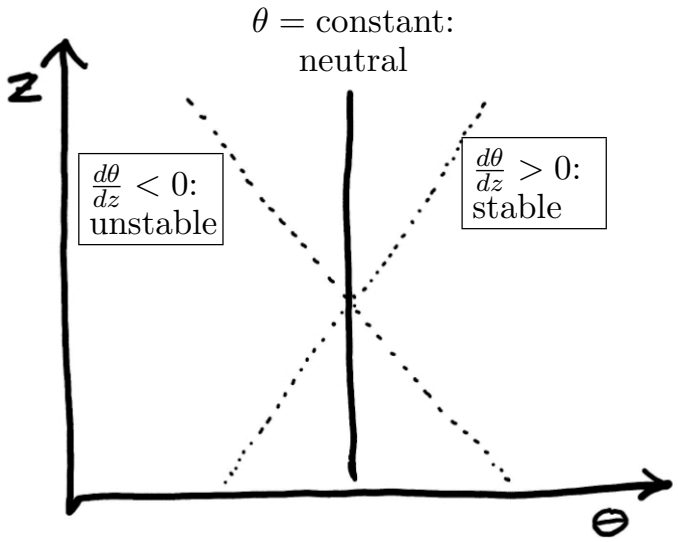


Figure 3.2: Schematic showing the convective lapse rate in potential temperature ($d\theta/dz = 0$) as well as convectively stable ($d\theta/dz > 0$) and unstable ($d\theta/dz < 0$) profiles. Figure courtesy Adam Showman.

much more intuitive way to diagnose whether an atmosphere is statically stable[17], and so I recommend using it where possible.

3.1.4 Convective available potential energy (CAPE)

Convective Available Potential Energy (CAPE) is a useful measure to determine the susceptibility of a given temperature (and moisture, but so far we consider dry cases alone) profile to deep convection (see Section 9.5.1 of Holton & Hakim, 2013). Specifically, CAPE is a measure of the maximum possible kinetic energy that a statistically unstable parcel can acquire, assuming no mixing with the environment and instantaneous adjustment to the local environmental pressure.

We can derive CAPE as follows. First, we write the acceleration of a parcel due to buoyancy (b) where the density of a parcel is ρ_p and that of the environment is ρ_e,

$$b = g\frac{\rho_e - \rho_p}{\rho_p} = g\frac{1/T_e - 1/T_p}{1/T_p} = g\frac{T_p - T_e}{T_e}, \tag{3.6}$$

[17]Potential temperature is also often the thermodynamic quantity used in 3D circulation models of planetary atmospheres. It's especially helpful that the primitive equations of meteorology map onto the shallow-water equations in isentropic coordinates...but that's for much later!

where T_p is the temperature of the parcel and T_e is the temperature of the environment. We can then calculate the maximum kinetic energy of a parcel displaced vertically from the level of free convection (LFC, the vertical point where deep convection begins, see Figure 3.3) to the level of neutral buoyancy (LNB, the higher level above the LFC where convection is inhibited) as

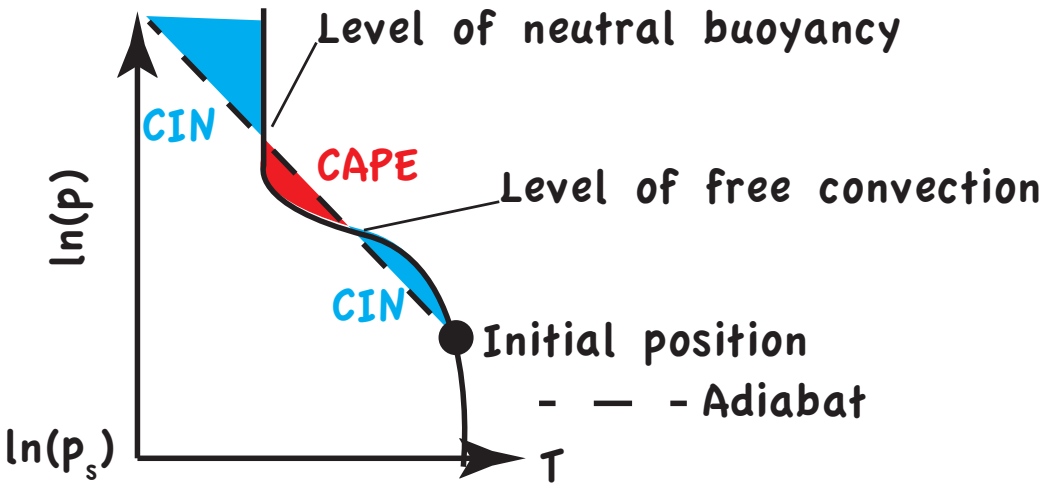


Figure 3.3: Schematic showing the trajectory of a parcel in T-ln(p) space alongside the LFC, LNB, and regions where CAPE is positive (red) and negative (blue). Figure courtesy Ray Pierrehumbert.

$$\frac{w_{\max}^2}{2} = \int_{z_{LFC}}^{z_{LNB}} g\left(\frac{T_p - T_e}{T_e}\right) dz = \int_{p_{LNB}}^{p_{LFC}} R(T_p - T_e) d\ln p, \tag{3.7}$$

where the second equality uses the ideal gas law and hydrostatic balance. This is the definition of CAPE – thus, we can express CAPE as

$$\text{CAPE} \equiv \int_{p_{LNB}}^{p_{LFC}} R(T_p - T_e) d\ln p. \tag{3.8}$$

CAPE is thus positive in regions that are convective, and negative (termed convective inhibition, CIN) in regions in which convection is inhibited.